Ruin, Drawdowns and the Kelly Criterion

Pass probabilities under prop-firm loss limits, and optimal stakes when the edge is uncertain

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30 October 2026

Abstract

A trader taking a funded-account ("prop-firm") challenge must reach a profit target before breaching an overall loss limit and a daily loss limit. I derive the classical gambler's-ruin probability and expected duration by solving a linear recurrence, extend the model to unequal wins and losses and to a daily limit with an absorbing Markov chain, and confirm every result by Monte Carlo simulation. I then derive the Kelly stake, which maximises long-run growth when profits compound, and show how uncertainty about the win rate changes the answer. Using a personal record of 5 wins in 12 trades with an average win 2.28 times the average loss, the pass probability at 1% risk per trade is 96% if that record reflects the true edge, but ranges from 2% to almost 100% across the 95% interval for the true win rate. Twelve trades cannot tell which.

Note on authorship. The derivations, code and text of this first version were produced by an AI assistant (Claude) at my request, checked by running the code and its tests, and reviewed by a second, separate AI review pass. The input record is my own. I am working through the derivations to reproduce them independently; the repository's development log records what was done, by whom and when.

1 Model

Each trade wins with probability p and loses with probability $q = 1 - p$, independently. A loss costs a fixed fraction r of the *initial* balance (the *risk per trade*); a win pays br, where b is the payoff ratio (average win divided by average loss). Profit and loss (P&L) is measured as a fraction of the initial balance. The account *passes* when P&L reaches the target T and *fails* when it reaches $-L$, or when the P&L of a single day reaches $-L_d$ (measured from the balance at the start of the day). Touching a limit counts as reaching it. The trader makes k trades a day. Throughout, $T = L = 10\%$, $L_d = 5\%$ and $k = 3$, typical of a first-phase challenge.

The expected P&L per trade, in units of r, is the *expectancy*

$$E = pb - q,$$

and the break-even win rate is $p = 1/(1 + b)$; for $b = 2.28$ it is 30.5%.

2 Equal wins and losses: the gambler's ruin

Take $b = 1$ and no daily limit. Measure P&L in steps of r and shift it so the loss limit is 0: the account starts at $i = L/r$ and passes at $N = (L + T)/r$, both assumed to be integers.

Proposition 1. *Let P_i be the probability of reaching N before 0 from i, and $\rho = q/p$. Then*

$$P_i = \begin{cases} \dfrac{1-\rho^i}{1-\rho^N}, & p \neq q, \\[2ex] \dfrac{i}{N}, & p = q. \end{cases}$$

Proof. Condition on the first trade. With probability p the account moves to $i+1$, with probability q to $i-1$, and from there the process starts afresh (the Markov property). Hence

$$P_i = p\,P_{i+1} + q\,P_{i-1}, \qquad 0 < i < N, \qquad P_0 = 0, \quad P_N = 1. \tag{1}$$

This is a linear recurrence with constant coefficients; with its two boundary values it has exactly one solution, since P_0 and P_1 determine every later term and P_1 is then fixed by $P_N = 1$. Trying $P_i = \lambda^i$ gives the characteristic equation $p\lambda^2 - \lambda + q = 0$, whose roots are

$$\lambda = \frac{1 \pm \sqrt{1-4pq}}{2p} = \frac{1 \pm |p-q|}{2p},$$

because $1 - 4pq = (p+q)^2 - 4pq = (p-q)^2$. So the roots are 1 and $q/p = \rho$.

If $p \neq q$ the roots differ and the general solution is $P_i = A + B\rho^i$. The boundary conditions give $A + B = 0$ and $A + B\rho^N = 1$, so $B = -1/(1-\rho^N)$ and $A = 1/(1-\rho^N)$, which is the first case.

If $p = q = \frac{1}{2}$ the root 1 is repeated and the general solution is $P_i = A + Bi$; the boundaries give $A = 0$ and $B = 1/N$. □

Proposition 2. *The expected number of trades until the account passes or fails is*

$$D_i = \begin{cases} \dfrac{i - NP_i}{q-p}, & p \neq q, \\[2ex] i(N-i), & p = q. \end{cases}$$

Proof. The game ends in finite expected time: in every block of N trades, N straight wins end it, which has probability $p^N > 0$, so the duration is at most a geometric number of blocks. Conditioning on the first trade again, and counting that trade,

$$D_i = 1 + p\,D_{i+1} + q\,D_{i-1}, \qquad D_0 = D_N = 0.$$

The homogeneous part is the recurrence (1), with solutions $A + B\rho^i$. For a particular solution of the full equation try $D_i = c\,i$: then $c\,i = 1 + p\,c(i+1) + q\,c(i-1) = 1 + c\,i + c(p-q)$, so $c = 1/(q-p)$. Therefore $D_i = i/(q-p) + A + B\rho^i$. The boundary conditions give $A = -B$ and $B(1-\rho^N) = N/(q-p)$, so

$$D_i = \frac{i}{q-p} - \frac{N}{q-p} \cdot \frac{1-\rho^i}{1-\rho^N} = \frac{i - NP_i}{q-p}.$$

When $p = q$, the particular solution is instead $D_i = -i^2$ (check: $-i^2 = 1 - \frac{1}{2}(i+1)^2 - \frac{1}{2}(i-1)^2 = -i^2$), and the boundaries give $D_i = i(N-i)$. □

For $r = 1\%$ the challenge is $i = 10$, $N = 20$. A fair game ($p = \frac{1}{2}$) passes with probability $\frac{1}{2}$ after 100 trades on average. A 5-point edge ($p = 0.55$) raises this to 88.1% after 76.3 trades; a 5-point deficit ($p = 0.45$) lowers it to 11.9%.

3 Unequal payoffs and a daily limit: an absorbing Markov chain

With $b \neq 1$ the recurrence has unequal steps, and the daily limit makes the outcome depend on the order of trades within a day, so I use a Markov chain instead.

States. Replace b by the nearest fraction w/m with denominator at most 50 (for $b = 2.28$ this is exact: 57/25) and count P&L in units $u = r/m$. A loss is then $-m$ units and a win $+w$ units, so every reachable P&L is a whole number of units. The state is the P&L at the start of a day, between the two limits.

Transitions. For each start state, list all 2^k win/loss sequences of a day, each with probability $p^{\text{wins}}q^{\text{losses}}$, and follow the P&L trade by trade. A sequence is *absorbed* as soon as the P&L reaches the target (pass), the overall limit (fail) or the daily limit relative to the day's start (fail); otherwise it ends the day in a new state. This gives a matrix Q of transitions between non-absorbed states and, for each outcome o, a vector R_o of one-day absorption probabilities.

Absorption. Let $h_o(s)$ be the probability that an account starting the day in state s eventually ends in outcome o. Conditioning on the first day, exactly as in Proposition 1,

$$h_o = R_o + Q\,h_o \quad \Longleftrightarrow \quad (I - Q)\,h_o = R_o,$$

and the expected number of days t satisfies $(I - Q)\,t = \mathbf{1}$. The matrix $M = (I - Q)^{-1} = I + Q + Q^2 + \cdots$ is the *fundamental matrix* of the chain (Kemeny and Snell, 1960): M_{ij} is the expected number of days spent in state j starting from i. It exists because absorption is reachable from every state (a run of wins always passes), so $Q^n \to 0$. The one exception is a rule that forbids every trade, which the code detects and reports. In practice the sparse linear systems are solved directly rather than inverting Q. Probabilities within a time limit come from multiplying the start distribution by Q once per day.

Validation. With $b = 1$, $k = 1$ and no daily limit, the chain reproduces Propositions 1 and 2 to at least six decimal places for every case tested, and a Monte Carlo simulation of 200,000 accounts per case, which uses the payoff as given rather than a lattice, agrees within sampling error (Table 1).

Table 1: Equal wins and losses, target and loss limit both 10%. Selected rows; all eight are in the repository.

r	p	Formula	Chain	Simulation (s.e.)	Formula D	Chain D
1%	0.40	0.017046	0.017046	0.0169 (0.0003)	48.30	48.30
1%	0.45	0.118501	0.118501	0.1189 (0.0007)	76.30	76.30
1%	0.50	0.500000	0.500000	0.5008 (0.0011)	100.00	100.00
1%	0.55	0.881499	0.881499	0.8811 (0.0007)	76.30	76.30
2%	0.40	0.116364	0.116364	0.1158 (0.0007)	19.18	19.18
2%	0.55	0.731717	0.731717	0.7316 (0.0010)	23.17	23.17

4 The challenge with the record's edge

My own record is 5 wins in 12 trades ($p = 41.7\%$) with $b = 2.28$, an expectancy of $+0.37$ of the amount risked per trade *if* the record is representative. Table 2 gives the exact outcome probabilities.

Three features stand out.

1. **The daily limit dominates above 1.5% risk.** Three losses cost $3r$, which reaches 5% once $r \geq 5/3\% \approx 1.67\%$; of the risks tested, 2% is the first above that. The pass probability drops from 90% at 1.5% to 53% at 2%. At $r = 2\%$ the overall limit can no

Table 2: Challenge outcomes for $p = 5/12$, $b = 2.28$, three trades a day.

Risk r	Pass	Fail (overall)	Fail (daily)	Expected days	Pass within 60 days
0.25%	100.0%	0.0%	0.0%	37.5	91.6%
0.5%	99.8%	0.2%	0.0%	19.3	99.2%
1%	96.1%	3.9%	0.0%	9.5	96.1%
1.5%	89.6%	10.4%	0.0%	5.8	89.6%
2%	53.0%	0.0%	47.0%	2.4	53.0%
3%	52.6%	0.0%	47.4%	1.4	52.6%
5%	41.7%	0.0%	58.3%	1.0	41.7%

longer be reached at all: a day with at most two losses ends at least $-4\%+4.56\% = +0.56\%$ up, so every day either gains or breaches the daily limit. A trader who skips any trade whose loss would breach the daily limit raises the pass probability at 2% risk from 53% to 83%.

2. **With a positive edge, smaller stakes pass more often but take longer.** Without a time limit, $r = 0.25\%$ passes almost surely but needs 37.5 days on average; within 60 trading days $r = 0.5\%$ does best of the risks tested (99.2%). This is the law of large numbers: more, smaller trades let the positive expectancy dominate the noise.

3. **With a negative edge the opposite tends to hold.** For $p = 0.25$ (expectancy -0.18) the pass probability rises from 0.1% at $r = 0.25\%$ to 25% at $r = 5\%$, though not monotonically (22% at 1.5%, 15% at 2%) because the daily limit interferes. When the game is unfavourable, fewer and larger bets give the negative expectancy fewer chances to act. This is consistent with the intuition behind Dubins and Savage's (1965) "bold play", although their theorem concerns a gambler who may change the stake after every bet, which this fixed-stake model does not allow.

How much the answer depends on the win rate. Twelve trades give a 95% (Wilson) interval for the true win rate of 19.3% to 68.0%. Across that interval the pass probability at 1% risk runs from 2.0% to almost 100% (Figure 1); even the break-even win rate of 30.5% lies inside it. The record cannot tell a near-certain pass from a near-certain failure.

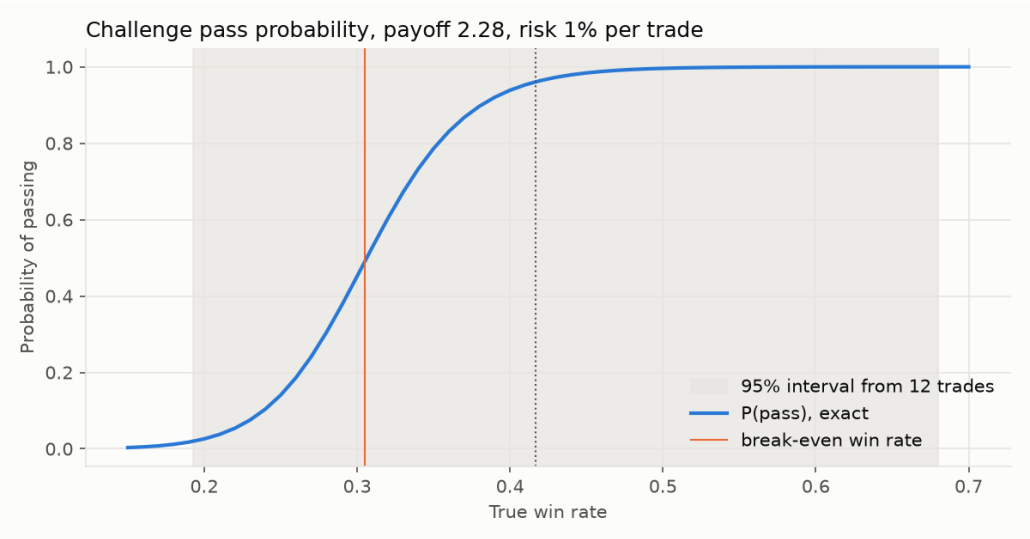


Figure 1: Exact pass probability against the true win rate, $b = 2.28$, $r = 1\%$.

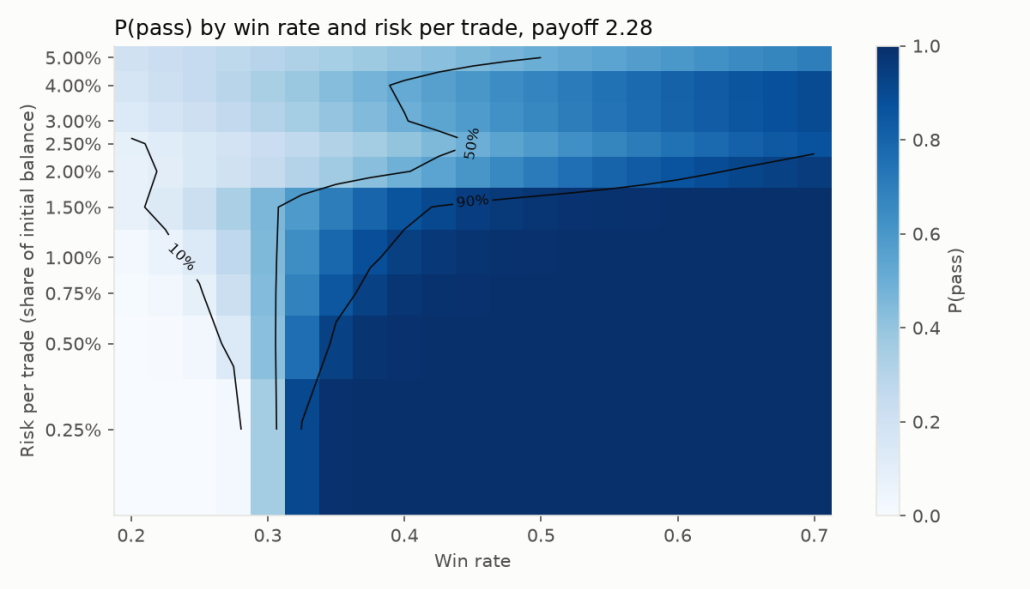


Figure 2: Pass probability by win rate and risk per trade. The step between 1.5% and 2% is the daily limit starting to bind.

5 The Kelly criterion

The challenge risks a fixed share of the *initial* balance. A trader whose profits compound instead risks a fraction f of the *current* balance. After n trades with W wins the balance is multiplied by $(1+bf)^W(1-f)^{n-W}$, so the growth rate per trade is $\frac{1}{n}\ln\big[(1+bf)^W(1-f)^{n-W}\big]$, which converges to

$$g(f) = p\ln(1+bf) + q\ln(1-f)$$

by the law of large numbers ($W/n \to p$). Kelly (1956) proposed maximising g; Breiman (1961) proved that the strategy maximising g eventually has more wealth than any essentially different strategy, with probability one.

Proposition 3. *For $0 \le f < 1$, g is strictly concave and is maximised at*

$$f^* = p - \frac{q}{b},$$

provided $f^* > 0$. If $f^* \le 0$ (zero or negative expectancy) the best stake is 0.

Proof. Differentiating,

$$g'(f) = \frac{pb}{1+bf} - \frac{q}{1-f}, \qquad g''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{q}{(1-f)^2} < 0,$$

so g is strictly concave and any stationary point is the global maximum. Setting $g'(f) = 0$: $pb(1-f) = q(1+bf)$, so $pb - q = bf(p+q) = bf$, giving $f^* = (pb-q)/b = p - q/b$. Since $g'(0) = pb - q$, the function is decreasing from $f = 0$ when the expectancy is not positive, and the constrained optimum is then $f = 0$. □

For the record, $f^* = 5/12 - (7/12)/2.28 = 16.1\%$ of the balance per trade. Across the 95% interval for p it runs from -16.1% (do not trade) to 54.0% (Figure 3).

Fractional Kelly. Using $\ln(1+x) \approx x - x^2/2$ for small f,

$$g(f) \approx f\mu - \tfrac{1}{2} f^2 m_2, \qquad \mu = pb - q, \quad m_2 = pb^2 + q,$$

where μ and m_2 are the mean and second moment of a trade's return per unit staked. This is a parabola with maximum at μ/m_2, a rough approximation to f^* (0.133 against the exact 0.161 for the record). Then $g(cf^*) \approx g(f^*)\,(2c - c^2)$: half Kelly keeps about 75% of the growth, and twice Kelly gives about zero growth. The exact values for the record (76.2% at half Kelly, 10.1% at twice Kelly) are close to these approximations even though f^* is not small.

Drawdowns. Simulating 20,000 paths of 500 trades at the record's edge, the balance fell at least 50% below its peak on 24% of paths at quarter Kelly, 99% at half Kelly and 100% at full Kelly; it fell 80% below its peak on 0.05%, 17% and 99% of paths. Full Kelly maximises growth, not comfort.

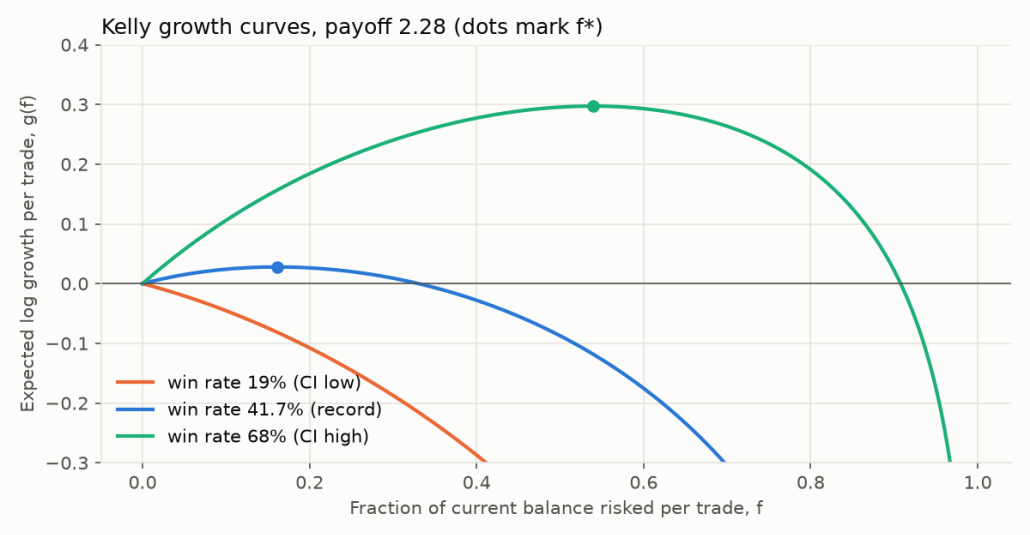


Figure 3: $g(f)$ for $b = 2.28$ at the record's win rate and at both ends of its 95% interval.

6 Kelly with an estimated win rate

In practice p is unknown. Suppose a trader estimates it from n trades as $\hat{p} = W/n$ and then stakes $c\,(\hat{p} - (1-\hat{p})/b)$, set to zero if negative. Because W is binomial, the expected true growth is an exact finite sum,

$$\mathbb{E}[g] = \sum_{w=0}^{n} \binom{n}{w} p^w q^{n-w}\, g\Big(\max\big\{0,\; c\,\big(\tfrac{w}{n} - \tfrac{1-w/n}{b}\big)\big\}\Big),$$

with g evaluated at the true p.

Growth is shown as a share of the growth available to someone who knows p. With 12 trades, full Kelly on the estimate gives *negative* expected growth. In 19% of samples (7 or more wins out of 12) the estimated stake is at least 40%, more than twice f^*, where the true growth is negative; because g is concave and falls steeply beyond f^*, those over-bets cost more than the under-bets in other samples gain. Betting half the estimated stake recovers 55% of the attainable growth. By 100 trades the penalty for full Kelly is small (82% of the attainable growth against 85% at the best multiple), and by 1,000 it disappears.

All stakes in this calculation are capped at 99%. Without the cap, full Kelly would stake everything after a sample of all wins ($\hat{p} = 1$), an event of probability $p^n > 0$, and the expected

Table 3: Best multiple c of the estimated Kelly stake, true $p = 5/12$, $b = 2.28$ known.

Trades n	Best c	Growth at best c	Growth at $c = 1$	P(growth < 0) at $c = 1$
12	0.50	55%	−9%	19%
30	0.65	67%	49%	7%
100	0.85	85%	82%	1%
300	0.95	94%	94%	0%
1000	1.00	98%	98%	0%

growth would be $-\infty$ for every finite n. This calculation also treats b as known; in reality it is estimated from the same few trades, which makes the case for a fraction of Kelly stronger still.

A subtle point: if instead one places a Bayesian prior on p and maximises the *posterior expected* growth, the answer is simply Kelly at the posterior mean, because g is linear in p. Uncertainty in p alone therefore does not shrink the Bayes-optimal stake; what makes plugging in a point estimate dangerous is the sampling error of that estimate together with the asymmetry of g.

7 Limitations

- Trades are independent with fixed win and loss sizes. Real outcomes vary in size, cluster in time, and may depend on market conditions.
- The record of 12 trades is self-reported and unverified; all numbers that use it are conditional on it.
- Prop-firm rules differ. Here the overall limit is static, the daily limit is measured from the start-of-day balance, losses of exactly the limit count as breaches, and the account passes the moment the target is reached; some firms use trailing limits, minimum trading days or equity-based measurement.
- The Kelly analysis ignores costs, and the drawdown results are simulations with a fixed horizon.

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